Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Consider the nonlinear equation $y = f(x) = x^3 + 1$, where the input x is a Gaussian random variable having mean of 1 and variance of 1. We desire to calculate the statistics of the output variable y using the EKF approach.

1 / 1 point

If we approximate $\mathbb{E}[f(x)]$ with $f(\mathbb{E}[x])$, what value to we find for $\mathbb{E}[y]$? Enter your answer with one digit to the right of the decimal point.

2

Correct
Yes. That is the right answer. Note, however, that the true expected value of y is actually 5. The EKF approach fails!

2. For the same problem described in question 1, if we desire to calculate the covariance of y using the EKF approach, we must first linearize $f(x)$. What is the value of $\hat{A} = \partial f(x)/\partial x$, evaluated at $x = \bar{x}$? Enter your answer with one digit to the right of the decimal point.

1 / 1 point

3

Correct
Yes. That is the correct result.

3. For the same problem described in question 1, we would calculate $\Sigma_{\tilde{y}} = \hat{A}\Sigma_{\tilde{x}}\hat{A}^T$ if we were using the EKF approach. What is the value you compute for $\Sigma_{\tilde{y}}$ using this method? Enter your answer with one digit to the right of the decimal point.

1 / 1 point

9

Correct
Yes. That is the correct answer. Note, however, that the true value is actually 60. The EKF approach fails!

4. For the same problem described in question 1, we now desire to calculate the statistics of the output variable y using the sigma-point approach, where we use the CDKF parameter values when doing so and assume that $h = \sqrt{3}$. What is the mean $\hat{y}$? Enter your answer with one digit to the right of the decimal point.

1 / 1 point

5

Correct
Yes. That is the correct answer. Note that this turns out to be the exact analytic result in this case as well!

5. For the same problem described in question 1, what is the value you compute for $\Sigma_{\tilde{y}}$ using the sigma-point approach? Enter your answer with one digit to the right of the decimal point.

1 / 1 point

54

Correct
Yes. That is the correct answer. Note that the analytic result is 60, so this is not perfectly true (but it is better than the EKF result).

6. Suppose that we store the state-prediction sigma points in matrix form as

1 / 1 point

$$\mathcal{X}_k^{x,-} = \begin{bmatrix} 0.6 & 0.3 & 0.8 & 0.9 & 0.4 \\ 0 & 0.6 & 0.8 & 0.2 & 0.2 \end{bmatrix}$$

and that $\alpha_k^{(\mathrm{m})} = 0.2$ for all k. We can compute that

$$\hat{x}_k^- = \begin{bmatrix} 0.6 \\ a \end{bmatrix}.$$

What is a? Enter your answer with two digits to the right of the decimal point.

0.36

Correct
Yes. That is the correct answer.

7. Suppose that we store the state-prediction sigma points in matrix form as

1 / 1 point

$$\mathcal{X}_k^{x,-} = \begin{bmatrix} 0.6 & 0.3 & 0.8 & 0.9 & 0.4 \\ 0 & 0.6 & 0.8 & 0.2 & 0.2 \end{bmatrix}$$

and that $\alpha_k^{(\mathrm{c})} = 0.2$ for all k. We can compute that

$$\Sigma_{\tilde{x},k}^- = \begin{bmatrix} 0.052 & 0 \\ 0 & b \end{bmatrix}.$$

What is b? Enter your answer with four digits to the right of the decimal point.

0.0864

Correct
Yes. That is the correct answer.

8. For the remaining questions in this quiz, you will execute the Jupyter Lab code for Lesson 3.2.5 and enter some computed results.

1 / 1 point

Execute the code as is. What is the rms error for the **velocity** state? Enter your answer with four digits to the right of the decimal point. Note that the code is originally configured to compute the rms error for all the states (position and velocity) so you will need to make minor modifications to compute the desired result.

0.0035

Correct
Yes. That is correct.

9. What is the maximum absolute estimation error for the **position** state? Enter your answer with three digits to the right of the decimal point.

1 / 1 point

0.010

Correct
Yes. That is correct.

10. What is the percentage of time that the **velocity** estimate is outside of its estimation confidence bounds? Enter your answer with one digit to the right of the decimal point.

1 / 1 point

0

Correct
Yes, that is correct.